

# Nevada County Experience

Visitor planning platform for **Western Nevada County** — feature overview & demonstration script

## What this is

A purpose-built tourism platform for **Western Nevada County** — the Gold Country foothills region centered on Nevada City and Grass Valley, with full coverage of Penn Valley, North San Juan, Rough & Ready, Washington, Chicago Park, Smartsville, and adjacent Colfax. (Truckee and the Sierra-side areas of Nevada County are out of scope for this build.)

The platform combines **161 curated experiences** with **460+ live-scraped events** from KVMR, Eventbrite, the Nevada City Chamber, the Grass Valley Chamber, and other local sources. Visitors discover what to do; the chamber team curates and approves what they see; the system runs without ongoing developer involvement.

|                        |                                                                                                           |
|------------------------|-----------------------------------------------------------------------------------------------------------|
| Geographic scope       | Western Nevada County (Gold Country foothills) + adjacent Colfax                                          |
| Total experiences      | 161 across 11 communities                                                                                 |
| Live event sources     | 6 — KVMR, Eventbrite, NC Chamber, GV Chamber, The Union, Go Nevada                                        |
| Event count (approved) | 460+                                                                                                      |
| Discovery vibes        | 9 themed + photo cards                                                                                    |
| Filter dimensions      | Area, Category (with sub-pills), Vibe, Date range                                                         |
| Tag taxonomy           | ~30 unique tags                                                                                           |
| Privacy posture        | No tracking · no analytics · no cookies. Itinerary save is opt-in only with explicit consent banner. Visi |
| Hosting requirement    | Any Linux/Windows server with Python; no database                                                         |
| Recurring monthly cost | \$5–\$20 typical (hosting + ~\$0.50/mo AI). See Cost Breakdown for detail.                                |
| First-year total       | ~\$80–\$150 typical. ~\$0 possible on free hosting tiers.                                                 |

## How to use this document

Pages 2–4 define *who* the platform serves — nine distinct visitor personas. Pages 5–6 list every feature, organized by audience. Pages 7–8 walk through an eight-minute demo, click by click. The remaining pages cover talking points, costs, and server migration.

# Who This Serves

Nine distinct visitor personas — each searches differently, measures success differently, and converts on different content.

A tourism platform either serves everyone weakly or serves a focused set of personas exceptionally well. Understanding who you're building for shapes navigation, content priorities, and where the chamber spends marketing budget. The nine personas below split into two macro-modes:

| Macro-mode                            | How they use the site                                                                              |
|---------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Browse-first</b> (vibe-shopping)   | Visitor is shopping a feeling — "romantic weekend," "active trip," "decompression." They want vibe |
| <b>Date-first</b> (calendar-shopping) | Visitor is shopping a calendar window — "what's happening that weekend." They want fresh event     |

## The nine personas

| Persona                            | Who & trip                                                                 | How they search differently                                                      |
|------------------------------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>1. Romantic Weekender</b>       | Couples 35-65, Bay Area / Sac, 1-3 nights, splurge on                      | "Where should we have the perfect date weekend?"</i> Shopping a feeling — the    |
| <b>2. Festival Pilgrim</b>         | Any age, anchored to a specific event/dater for X.                         | "What else should I do?"</i> Date-locked — searches the event                    |
| <b>3. Trail & Lake Seeker</b>      | 25-45, sporty, often gear-heavy, day/weekend nights                        | "Where can I hike / bike / swim / paddle this weekend?"</i> Needs physical spe   |
| <b>4. Multi-Gen Family</b>         | Parents 30-55 with kids 5-15, sometimes grandkids                          | "Will my kid parents enjoy this?"</i> Every venue gets the "will my 8-year-old   |
| <b>5. Arts & Heritage Traveler</b> | 40-70, college-educated, often retired                                     | "What's the depth here?"</i> Wants substance over photogenic stops. Reads        |
| <b>6. Wine Country Foodie</b>      | 30-60, food/wine enthusiasts, often wine alternates typically              | "Where are the best things to eat?"</i> Meal-anchored trip planning. Needs prove |
| <b>7. Wellness Refugee</b>         | 30-55, urban professional, burnt out, wife has to                          | "Will I actually be able to slow down here?"</i> Filtering for what's NOT there  |
| <b>8. Local "What's Up"</b>        | Nevada County residents themselves/what's going on                         | "What's going on tonight?"</i> Already knows the venues. Wants pure events       |
| <b>9. Maker Traveler</b>           | 25-65, professional or semi-retired, friends to forge ahead if it's bagged | "Where can I find a workshop / throw a pot."</i> Shopping a SKILL, not a         |

■ **Maker Traveler highlight:** Western Nevada County is unusually well-positioned here — The Curious Forge, Wolf Craft School, ASiF, and Crafffolk Collective form a maker ecosystem few foothills counties can match. This persona has the highest revenue-per-visit (2-3 nights, workshop fee, 4-6 meals out, lodging usually unbooked from comparison sites), and the marketing story writes itself.

# How well the current site serves each persona

| Persona                  | Today      | Gap / next step                                                                                 |
|--------------------------|------------|-------------------------------------------------------------------------------------------------|
| Romantic Weekender       | ■ Strong   | Relaxed vibe, B&B sub-pill, real B&B photos already in place                                    |
| Festival Pilgrim         | ■■ Partial | Manual filtering required. Seasonal/event-anchored itineraries (planned) directly fixes this.   |
| Trail & Lake Seeker      | ■ Strong   | Hiking/Biking/Swimming/Boating/Fishing/Running tags + sub-pills shipped recently. Difficulty    |
| Multi-Gen Family         | ■■ Partial | Family vibe exists. Age-range guidance per entry isn't structured yet.                          |
| Arts & Heritage Traveler | ■ Strong   | Historic and Arts vibes both rich. Long-form storytelling is the next polish.                   |
| Wine Country Foodie      | ■ Strong   | Foodie vibe + Restaurant sub-pills + 9 wineries + 9 tasting rooms in db.                        |
| Wellness Refugee         | ■ Strong   | Recently broadened (Restorative tag, 5 new massage venues, float/spa/yoga properly captu        |
| Local "What's Up"        | ■■ Weak    | Same filtering as visitors today. A "this week" landing page + email digest would build residen |
| <b>Maker Traveler</b>    | ■■ Partial | <b>Hands-On vibe + Workshops sub-pill in place</b>, but workshop calendars (Curious For         |

## Strategic recommendation

Of the nine personas, the chamber should **over-serve four** in site content, marketing, and budget. The remaining five should be served gracefully by the platform but not be primary marketing targets.

| <b>Over-serve (focused marketing)</b><b>Why</b> |                                                                                                       |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| ■ Maker Traveler                                | Highest revenue per visit; defensible market position; The Curious Forge alone is a destination       |
| Festival Pilgrim                                | Drives concentrated peak-season tourism (Cornish Christmas, Wild & Scenic, Bluegrass).                |
| Romantic Weekender                              | Bay Area weekend market is enormous and well-suited to the foothills product. Highest per-cap         |
| Wine Country Foodie                             | Napa-alternative positioning is achievable. Sierra foothills wine region needs visibility this site c |

The other five personas (Trail & Lake, Family, Arts & Heritage, Wellness, Local) are well-served by what's built and don't require concentrated marketing — they self-discover or are served by adjacent channels (REI events, parent groups, local newspapers).

## What the platform does for each persona

A single visit to the home page should let any of the nine personas find their answer in under 30 seconds. The discovery paths are:

- **Vibe cards** (9 themed) — direct match for Romantic, Active, Foodie, Wellness, Hands-On (Maker), Family, Festivals, Arts, Historic
- **Category dropdown + sub-pills** — matches specific shopping intent (Lodging → B&Bs, Outdoor → Swimming, etc.)
- **Date filter** — Festival Pilgrim and Local "What's Up" go straight here
- **Smart Suggestions** in the itinerary modal — works for any persona once they've added 2+ items

- **Sub-pills** like ■ Workshops & Classes (Activity) or ■ Spa & Float (Wellness) — fine-grained intent

*"You don't need to serve everyone equally. You need to serve the four right personas exceptionally — and not actively annoy the other five. The platform already does the second; the marketing investment is what closes the first."*

# Feature Inventory

Everything the platform does today — built and tested.

## For Visitors

### Discovery & Filtering

| Feature             | Description                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------|
| Area filter         | Nevada City, Grass Valley, Penn Valley, North San Juan, Rough & Ready, Washington, Chicago P             |
| Category dropdown   | 15 categories — Lodging, Outdoor, Restaurant, Music venue, Wellness, etc.                                |
| Sub-pills           | 7 categories show contextual sub-types (e.g. Lodging → Hotels, B&Bs, Glamping, Campgrounds, R            |
| Themed vibes        | 9 photo cards: Historic, Arts & Music, Hands-On, Foodie, Active, Relaxed, Wellness, Family, Festiv       |
| Date filter         | Free range + quick presets: Today, This Weekend, This Week                                               |
| Auto-season hiding  | Visiting in November? Summer-only experiences disappear automatically                                    |
| Active filter chips | Each active filter shown as a chip — one-click clear                                                     |
| Tag-based search    | Six new activity tags (Hiking, Biking, Running, Swimming, Fishing, Boating) attached to every relevant e |

### Planning Tools

| Feature              | Description                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------|
| "Add to My List"     | Heart on every card; floating itinerary button shows running count                                          |
| My Itinerary modal   | Numbered stops, removable, with hours and direct links                                                      |
| ■ Opt-in device save | <b>Privacy-by-default.</b> On first add, visitor is asked: "Save your itinerary on this device?" Yes → su   |
| ■ Share / Save link  | One tap → native phone share sheet (text, email, AirDrop, copy link). The link restores the entire itinerar |
| Cross-device handoff | Build on phone in the morning, text the link to yourself, open on iPad in the car. No login, no sync servic |
| Map view             | Leaflet/OpenStreetMap with custom gold pin markers                                                          |
| Print itinerary      | One click — formatted for paper hand-off (also "Save as PDF" from print dialog)                             |
| ■ Smart Suggestions  | Algorithmic — finds nearby experiences and matching events; shows "0.3 mi away" labels                      |
| Events inline        | Upcoming events show alongside places, filtered by the same vibe and dates                                  |
| Google Maps pin      | Custom SVG drop-pin button on each card opens directions                                                    |
| RSS feed             | /feed.rss publishes approved events for partner sites and aggregators                                       |
| Mobile responsive    | All features work on phone & tablet                                                                         |

# For the Tourism Body / Admin

## Content Management

| Feature                     | Description                                                          |
|-----------------------------|----------------------------------------------------------------------|
| Inline experience editor    | Sortable table of all 161 entries — edit, add, delete in browser     |
| Tag taxonomy editor         | Add/remove/rename tags; changes propagate to filtering immediately   |
| Drag-drop itinerary builder | Three-panel UI: itineraries · stops · browse                         |
| Events Queue                | Pending / Approved / Dismissed with filter buttons                   |
| Approve all                 | One-click bulk approval of pending events; auto-dismisses past dates |
| Auto-prune                  | Past events removed nightly; dismissed items purged after 60 days    |
| Source management           | Add/edit/disable scraper source URLs without touching code           |

## Automation

| Feature               | Description                                                                           |
|-----------------------|---------------------------------------------------------------------------------------|
| 6 scrapers            | KVMR, Eventbrite Nevada, NC Chamber, GV Chamber, The Union (RSS first), Go Nevada     |
| Auto-tagging          | Keyword pipeline — every scraped event gets baseline tags                             |
| ■ AI Categorize       | Claude Haiku refines tags, infers area & venue, scores quality (~\$0.20 per full run) |
| On-demand scraping    | "Update Now" button fires the scrape pipeline; status bar shows progress              |
| Idempotent processing | AI never re-categorizes already-processed events; cheap on subsequent runs            |
| Graceful degradation  | Site works without the server (static fallback); without an API key (no AI)           |

## Architecture & Operations

| Feature           | Description                                                                |
|-------------------|----------------------------------------------------------------------------|
| No database       | JSON files — easy backup, version control, no DBA needed                   |
| Python & Flask    | Single small server; runs on any Linux/Windows VPS                         |
| No vendor lock-in | No SaaS subscription; full code, full data ownership                       |
| Self-hostable     | Works behind Apache/Nginx; integrates with existing chamber infrastructure |
| Static-friendly   | Public site can be served as plain HTML if API isn't needed                |
| Open standards    | OpenStreetMap, RSS 2.0, plain JSON — no proprietary formats                |

# Visitor-Centered Design Decisions

Why the platform is easy to use — deliberate UX choices, not accidents.

Most tourism sites lose visitors at predictable friction points: cluttered filtering, confusing day-by-day planning, lost itineraries, forced sign-ups, and clicking-out moments that never bring people back. This platform was built around explicit decisions to remove those friction points.

## How key decisions improve the visitor flow

| Design decision                                        | Why it matters                                                                                                    |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Day-by-day itinerary view</b>                       | A flat list of stops hides the question "where do I sleep tonight?" Visitors think in stops, not nights — and s   |
| <b>"Find Lodging" CTA at top of itinerary</b>          | When the visitor has empty nights, a gold banner appears at the top of My Itinerary with a one-click "Find        |
| <b>Find Lodging clears active vibes</b>                | Vibes (Foodie, Active, Wellness) describe a discovery mood — not lodging properties. When a visitor expl          |
| <b>Add Day returns to browse mode</b>                  | "Add another day" doesn't leave the visitor staring at an empty day inside the modal — it closes the modal        |
| <b>Events go into the itinerary, not clicking KVMR</b> | "Clicking KVMR" sent visitors off-site to KVMR/Eventbrite — many never came back. New flow lets visi              |
| <b>Calendar dates on day headers</b>                   | When visitor sets a trip-start date, abstract "Day 1 / Day 2" become concrete: "DAY 1 · Sat, May 17". Spec        |
| <b>Vibe-level pills for finer granularity</b>          | Picking the Foodie vibe alone is broad. Sub-pills under the vibe (■ Restaurants · ■ Wineries · ■ Markets ·        |
| <b>Direct × remove on cards</b>                        | No "Manage Mode" toggle. The × button is always visible on cards once added. Visitor stays in browse-flo          |
| <b>Opt-in itinerary save</b>                           | On first add, visitor is asked: "Save your itinerary on this device?" Yes → survives tab close + browser res      |
| <b>Type-representative event placeholder</b>           | Every event without a real photo gets a colored gradient with its type emoji — music events look different t      |
| <b>"All Future" date filter as default</b>             | Replaces the implicit "no buttons active = no filter" state with an explicit default. Visitors immediately see th |
| <b>Auto-clearing seasonal experiences</b>              | Visiting in May? Summer-only experiences hide automatically. The system knows seasons and respects th             |

## What ties these together

Every choice listed above came from asking: "where does the visitor get stuck or leak out?" — then removing the friction. Vibes don't fight category filters. The itinerary doesn't require accounts. Events don't hijack the visitor to KVMR. Days aren't abstract numbers. Manage mode doesn't exist because it doesn't need to.

The result: a planning workspace where visitors complete trips, not just browse them. The pixel-level decisions add up to the strategic outcome — overnight stays.

# Demonstration Script

Eight-minute live walkthrough — visitor experience first, admin second.

## Setup (before the meeting)

- Run `python server.py` in the project folder
- Open `http://localhost:5000` in a fresh browser window
- Confirm **My Itinerary** is empty (refresh / clear if needed)
- Have a second tab ready on the public view, but DON'T show admin until Act 4
- Confirm the date is set in 2026 so seasonal experiences behave correctly

## Act 1 — "I'm a visitor planning a trip" · 2 min

*"Pretend you're a couple from Sacramento planning a romantic November getaway."*

| Click                                       | What to say                                                                                                                      |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Click "This Weekend" date preset            | Notice the trail and outdoor experiences that are summer-only just disappeared. The system knows you're in the area.             |
| Click the Festivals & Celebrations category | Overland Christmas, Victorian Christmas, and Mardi Gras appear at the top — exactly the things you'd expect to find in the area. |
| Pause on a card                             | Point out: real photo (pulled from the property website), tags, hours, notes, the More Info link, and the heart icon.            |

## Act 2 — "Let me narrow further" · 90 sec

| Click                                    | What to say                                                                                 |
|------------------------------------------|---------------------------------------------------------------------------------------------|
| Clear date filter (X on the chip)        | Filter chips and dates clear instantly.                                                     |
| Click Relaxed vibe (alongside Festivals) | "You can layer multiple vibes. The grid recombines on the fly."                             |
| Category dropdown → Lodging              | Sub-pills appear: Hotels · B&Bs/Inns · Glamping · Campgrounds · RV Parks.                   |
| Click ■ B&Bs / Inns pill                 | "These are real Victorian B&Bs in Nevada City — Broad Street Inn, Flume's End, Piety Hill." |
| Add 2-3 favorites to the list            | The heart icon flips, the floating itinerary count increments.                              |



## Act 3 — "Building my weekend" · 2 min

| Click                                    | What to say                                                                                         |
|------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Clear the Lodging filter                 | Back to the full grid.                                                                              |
| Switch to Foodie vibe                    | Restaurants, wineries, breweries, and food festivals appear together.                               |
| Add Lola Restaurant + Three Forks Bakery | Map fills, count goes up.                                                                           |
| Switch to Historic & Gold Rush vibe      | Empire Mine, Narrow Gauge Railroad Museum surface to the top.                                       |
| Add Empire Mine to the list              |                                                                                                     |
| Click the My Itinerary floating button   | Modal opens. <b>This is the magic moment.</b>                                                       |
| Scroll to ■ More to Explore              | "The system noticed everything's clustered in Grass Valley + downtown Nevada City — so it's         |
| Click + Add to List on a suggestion      | Suggestions instantly recompute — list grows, new neighbors appear.                                 |
| Click ■ View on Map                      | All stops appear on a Leaflet/OSM map with custom numbered gold pins.                               |
| Add the first item (if fresh visit)      | <b>A small banner slides up:</b> "Save your itinerary on this device?" — explicit consent. <b></b>  |
| Close the map, click ■ Share / Save      | "Now imagine I built this on my phone — I want my partner to see it." On phone the native share     |
| Reopen the modal                         | "Notice the line at the bottom: <i>Saved on this device · Forget on this device</i> . Visitor can r |
| Click ■ Print                            | Print preview opens — "Or save as PDF and email it. Print it for the fridge."                       |

## Act 4 — "Behind the scenes" · 90 sec

*"Now let me show you what the chamber team would actually use."*

| Click                               | What to say                                                                                    |
|-------------------------------------|------------------------------------------------------------------------------------------------|
| Click ■ Manage Database (top right) | Admin panel opens. Three management tabs: Experiences · Itineraries · Events Queue · Scrap     |
| Stay on the Experiences tab         | "All 161 entries — inline editable. No developer needed. Add a new venue, rename it, retag it, |
| Click ■ Events Queue                | "Where chamber staff review what's scraped overnight."                                         |
| Show pending count + filter buttons | "KVMR posted twelve new events; click Approve All; done in five seconds."                      |
| Click ■ AI Categorize               | "For about 20 cents we can have Claude clean up area names, infer venues for KVMR's vague      |
| Click ■ Scraper Sources             | "Add a new feed — say SYRCL's events RSS — paste the URL, save. The next overnight run         |

## Act 5 — Close · 60 sec

*"What this gives Nevada County:"*

- **Time saved for visitors** — they find Cornish Christmas + a B&B in 30 seconds instead of 30 minutes across five different sites

- **Local business visibility** — every chamber member appears under multiple discovery paths: vibe, area, category, sub-pill, date filter, search-like tags
- **Privacy without compromise** — opt-in itinerary save with explicit consent, cross-device share via link, no tracking, no accounts, no emails. Visitors stay anonymous; the chamber gets credit for respecting them.
- **No vendor lock-in** — open files, no database, no SaaS contract
- **Self-publishing** — the public RSS feed is something other sites and apps can pick up automatically
- **Modern AI integration** — categorization today, smart visitor suggestions next, all opt-in and bounded by cost
- **Fresh data, low maintenance** — scrapers run on a schedule; chamber staff review and approve in minutes a week

# Talking Points

Anticipated questions and the right answers.

| Question                                                 | Answer                                                                                                                  |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Can we host this ourselves?                              | Yes — single Python file, runs on any Linux box. Apache or Nginx friendly. The code can sit in a Git repo.              |
| What if a venue closes or changes hours?                 | Manage Database → click the row → edit or delete. Saved instantly. No developer. No release process.                    |
| Can we add events from [specific source]?                | Yes. Each scraper is a 100-line Python file. New sources are added by writing one matching that pattern.                |
| How fresh is the event data?                             | Last scrape timestamp shows in the admin status bar. Schedule a daily cron job for hands-off freshness.                 |
| Is the AI tagging accurate?                              | Claude Haiku gets the obvious cases right (Center for the Arts → Grass Valley). Every AI field is adjustable.           |
| What does it cost monthly?                               | Typical: \$10-\$20/month all-in (hosting + AI). Lean setup possible at ~\$1/month using a free hosting tier.            |
| Can we white-label this for the chamber?                 | Yes. Colors, logo, copy, photos all live in the HTML/CSS. No core changes needed.                                       |
| Does it work without internet?                           | The static site degrades gracefully — visitors see curated experiences but not live events. Useful as a backup.         |
| Privacy / data collection?                               | No tracking, no analytics, no cookies by default. Visitor itineraries are <b>never</b> stored without explicit consent. |
| What if a visitor wants to save / share their itinerary? | Two independent options: (1) <b>Save on this device</b> — opt-in via consent banner; survives tab closure.              |
| Accessibility?                                           | Semantic HTML, alt text on photos, keyboard navigation. WCAG-aligned but not formally audited.                          |

## Watch out for during the demo

- **Server must be running** — if you forget, the Events tab and AI button will be empty. The site degrades gracefully but the demo is much weaker.
- **Don't open the admin panel first.** Visitors first; "and here's how easy it is to maintain" second.
- **Skip the feature-list slides.** The live demo IS the pitch. Use this PDF as your prep, not your slides.
- **Suggestions need 1+ items first.** Always add something before opening the My Itinerary modal.
- **If WiFi is bad,** the Leaflet map and any uncached photos may fail. Have a fallback screenshot.
- **Audience-fit:** the Foodie / Festivals / B&Bs combination is most likely to land — they're universally appealing categories.

## Leave-behind one-liner

*"This isn't a website — it's a planning workspace for visitors and a living directory for the chamber, with AI-assisted curation built in. Privacy-respecting by default, owned by the county outright, maintained in minutes per week."*

# Cost Breakdown

What it actually costs to run, based on real measured usage.

## Recurring monthly costs

| Line item                                      | Notes                                                                                                                           | Monthly cost              |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| Web hosting                                    | Any small Linux VPS (DigitalOcean, Linode, AWS Lightsail). Standard option available if no admin features needed. \$10-\$20/mo. | \$10 – \$20               |
| Domain name                                    | Optional — only if not already using existing chamber domain. Renewed annually at ~\$12-15/year.                                | \$12 – \$15               |
| SSL certificate                                | Free via Let's Encrypt; renews automatically.                                                                                   | \$0                       |
| AI event categorization                        | Anthropic Claude Haiku API. Per measured usage (~460 events) \$0.30-\$1.50 run, then \$0.01-\$0.02 per scheduled event.         | \$0.30 – \$1.50           |
| Email / user accounts                          | No user accounts in the system. No email service required.                                                                      | \$0                       |
| Database                                       | No database — JSON files. No DB hosting fees.                                                                                   | \$0                       |
| Analytics                                      | No third-party tracking by default. If desired, free options (Plausible self-hosted, GoatCounter) available.                    | \$0                       |
| <b>Realistic monthly total</b>                 |                                                                                                                                 | <b>\$5 – \$25</b>         |
| Bare-minimum monthly total (free hosting tier) |                                                                                                                                 | \$0.30 – \$1.00 (AI only) |
| Typical chamber operation (paid VPS + AI)      |                                                                                                                                 | \$10 – \$20               |

## One-time / annual costs

| Line item                  | Notes                                                                                                         | Cost        |
|----------------------------|---------------------------------------------------------------------------------------------------------------|-------------|
| Initial setup & deployment | Configure server, install Python, deploy code, set up daily scraper, one-time point domain. One person-hours. | one-time    |
| Domain renewal             | If purchasing new domain (e.g. visitwesternnevada.org).                                                       | ~\$12-15/yr |
| Content review (annual)    | Walking through all 161 experiences once a year to verify hours, staff, etc. Roughly 4-6 staff hours.         | \$200s/ann  |

## AI cost detail (the only variable expense)

The AI categorization cost is the only line item that scales with usage. Here's exactly what it costs based on measured token counts:

| Run pattern                                         | Per-run cost | Monthly equivalent |
|-----------------------------------------------------|--------------|--------------------|
| First-time bulk categorization (current 460 events) | \$0.29       | one-time           |

| Run pattern                                               | Per-run cost            | Monthly equivalent           |
|-----------------------------------------------------------|-------------------------|------------------------------|
| Daily scrape + categorize (~10-30 new events/day)         | \$0.01 - \$0.02 per run | \$0.30 – \$0.60 / mo         |
| Weekly scrape + categorize (~70-200 new events/week)      | \$0.04 - \$0.13 per run | \$0.16 – \$0.52 / mo         |
| Quarterly full re-categorize (with --force)               | \$0.30 per run          | \$0.10 / mo equiv.           |
| Realistic chamber operation (weekly + occasional re-runs) |                         | <b>~\$0.50 – \$1.00 / mo</b> |

## Built-in cost protection

- **Hard usage cap on Anthropic side** — recommended \$5/month spending limit set in API console. Cannot ever exceed it; emails sent at 50% / 75% / 100%.
- **Idempotent processing** — events that have already been categorized are skipped automatically. No accidental re-billing on the same data.
- **Optional, not required** — the entire AI feature can be turned off (no API key set) and the site still functions on raw scraper data.
- **Quality filter blocks waste** — events flagged "low quality" by AI are auto-hidden from the public site, reducing noise without manual review.

# Annual Cost Summary

Three realistic scenarios for the first year of operation.

## Three operating scenarios

| <b>Scenario</b>          | <b>Lean</b>                   | <b>Typical</b>           | <b>Premium</b>                 |
|--------------------------|-------------------------------|--------------------------|--------------------------------|
| Hosting                  | Free tier (e.g. Oracle Cloud) | \$5/mo VPS               | \$20/mo managed VPS            |
| Domain                   | Use existing chamber domain   | New domain (~\$15/yr)    | Premium domain                 |
| AI categorization        | Disabled or rare              | Weekly runs (~\$0.50/mo) | Daily runs + re-runs (~\$1/mo) |
| Content review           | 4 hrs / year staff time       | 6 hrs / year             | Quarterly review               |
| Backups                  | Manual quarterly              | Automated weekly         | Automated daily off-site       |
| <b>First-year total</b>  | <b>~\$0 – \$50</b>            | <b>~\$80 – \$150</b>     | <b>~\$300 – \$500</b>          |
| <b>Subsequent annual</b> | <b>~\$0 – \$20</b>            | <b>~\$70 – \$130</b>     | <b>~\$280 – \$480</b>          |

## What this means in practice

For comparison, a commercial off-the-shelf tourism platform (CrowdRiff, Simpleview Destination Stack, or similar) typically runs **\$5,000 – \$30,000 annually** with multi-year contracts and per-feature pricing. The Nevada County Experience build delivers comparable visitor-facing capability for one to three orders of magnitude less, and the chamber owns all the code and data.

## What's NOT a cost

- **Per-listing fees** — every chamber member can be listed at no incremental cost.
- **Per-event fees** — scraped events come in for free; AI categorization is the only variable cost.
- **Per-visitor fees** — no rate-based pricing; if traffic 10×'s tomorrow, hosting cost barely changes.
- **License fees** — no proprietary software. Python, Flask, Leaflet, OpenStreetMap, all open source.
- **Vendor termination fees** — owned outright. No vendor to leave.
- **Migration costs** — JSON files are portable; can switch hosts in an afternoon.
- **Privacy compliance overhead** — no analytics, cookies, or trackers means no GDPR/CCPA banners, no privacy review, no consent platform fees. Itinerary save is opt-in by design.

## Recommended starting budget

**\$200 in the first year** covers everything: domain (\$15), modest VPS hosting (\$60/year), AI categorization (\$10), and a small reserve. Subsequent years drop to ~\$85 if domain doesn't need renewal that year. No staff training cost — the admin panel is intentionally simple enough that anyone who can edit a spreadsheet can maintain it.

# Migration to a County Server

Step-by-step deployment from local development to production hosting.

This is the full sequence to move the platform from a developer laptop to a production server the chamber controls. Estimated effort: **one focused half-day** for someone comfortable with Linux, **two days** for someone learning as they go. No code changes are required — the same files run locally and in production.

## Phase 0 — Decisions before you start

Lock these down first. Each affects what comes next.

| Decision             | Recommended                                | Alternative                                            |
|----------------------|--------------------------------------------|--------------------------------------------------------|
| Hosting provider     | DigitalOcean (\$6/mo droplet) or Linode    | AWS Lightsail · chamber's existing host · county-IT VM |
| Operating system     | Ubuntu 24.04 LTS                           | Debian 12 · RHEL 9 · Windows Server (more work)        |
| Domain               | New: visitwesternnevada.org (~\$15/yr)     | Subdomain: experience.nevadacountychamber.com          |
| Server account owner | Chamber / county directly                  | Contracted dev (lock-out risk)                         |
| SSL certificate      | Free Let's Encrypt (auto-renew)            | Paid cert (no real benefit)                            |
| Backups              | Provider snapshots + monthly off-site copy | Snapshots only · weekly tar to chamber NAS             |

**Decision-point:** if the county already has a web server (Apache or nginx running on a county-IT box), you can run this *next to* the existing site. If not, a fresh small VPS is cleaner and the cost is trivial.

## Phase 1 — Pre-flight checklist

Confirm all of these before continuing:

- Domain registered (or subdomain delegated)
- DNS access confirmed — you can edit A records
- Anthropic API key created at console.anthropic.com (with \$5/mo cap set)
- SSH key generated locally (id\_ed25519 or id\_rsa)
- You can SSH and copy files between your laptop and the server
- Cloudflare or similar set up (optional — adds free DDoS + caching)

# Phase 2 — Provision the server

## 2a. Buy the droplet

DigitalOcean → Create → Droplet → **Ubuntu 24.04 LTS** → Basic plan, \$6/mo (1 GB RAM, 25 GB SSD) → datacenter near the West Coast (San Francisco 3) → add your SSH key → name it nevada-county-experience → Create.

## 2b. First login + lockdown

```
# From your laptop
ssh root@<droplet-ip>

# Create a non-root user, give them sudo
adduser ncexp
usermod -aG sudo ncexp
mkdir -p /home/ncexp/.ssh
cp ~/.ssh/authorized_keys /home/ncexp/.ssh/
chown -R ncexp:ncexp /home/ncexp/.ssh
chmod 700 /home/ncexp/.ssh
chmod 600 /home/ncexp/.ssh/authorized_keys

# Disable root SSH login
sed -i 's/PermitRootLogin yes/PermitRootLogin no/' /etc/ssh/sshd_config
sed -i 's/#PasswordAuthentication yes/PasswordAuthentication no/' /etc/ssh/sshd_config
systemctl restart ssh
exit

# From now on, SSH as ncexp
ssh ncexp@<droplet-ip>
```

## 2c. System packages

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y python3 python3-pip python3-venv \
    nginx git ufw certbot python3-certbot-nginx \
    chromium-browser chromium-chromedriver

# Firewall: SSH + HTTPS only
sudo ufw allow OpenSSH
sudo ufw allow 'Nginx Full'
sudo ufw enable
```



# Phase 3 — Deploy the application

## 3a. Get the code onto the server

**Recommended:** push to a private GitHub repo from your laptop, clone on the server. **Alternative:** zip the folder and use scp.

```
# On your laptop (one-time)
cd "C:\Users\Curious Forge\Documents\nevada-county-experience"
git init
git add .
git commit -m "Initial commit"
gh repo create nevada-county-experience --private --source=. --push

# On the server
cd /home/ncexp
git clone https://github.com/<your-user>/nevada-county-experience.git
cd nevada-county-experience
```

## 3b. Python environment

```
cd /home/ncexp/nevada-county-experience
python3 -m venv .venv
source .venv/bin/activate
pip install flask requests beautifulsoup4 lxml selenium \
        webdriver-manager python-dateutil reportlab
deactivate
```

## 3c. Create the secrets file

scraper/config.py is gitignored, so the keys never leave this server.

```
cat > scraper/config.py <<'EOF'
GOOGLE_PLACES_API_KEY = "<paste from your local config.py>"
ANTHROPIC_API_KEY = "sk-ant-api03-..."
CENTER_LAT = 39.2350
CENTER_LNG = -121.038
RADIUS_M = 35000
MIN_RATINGS = 5
MIN_RATING = 3.5
EOF
chmod 600 scraper/config.py
```

## 3d. Smoke test

```
source .venv/bin/activate
python server.py
# Should show "Running on http://127.0.0.1:5000"
# Press Ctrl+C
deactivate
```

## Phase 4 — Run as a persistent service

systemd auto-starts the server on reboot and auto-restarts if it crashes. Logs go to journalctl.

```
sudo tee /etc/systemd/system/ncexp.service > /dev/null <<'EOF'
[Unit]
Description=Nevada County Experience web app
After=network.target

[Service]
Type=simple
User=ncexp
WorkingDirectory=/home/ncexp/nevada-county-experience
Environment="PATH=/home/ncexp/nevada-county-experience/.venv/bin"
ExecStart=/home/ncexp/nevada-county-experience/.venv/bin/python \
    server.py --host 127.0.0.1 --port 5000
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
EOF

sudo systemctl daemon-reload
sudo systemctl enable ncexp
sudo systemctl start ncexp
sudo systemctl status ncexp      # should show "active (running)"

# View logs anytime:
journalctl -u ncexp -f
```

## Phase 5 — Reverse proxy + HTTPS

Flask shouldn't face the internet directly. nginx sits in front and terminates HTTPS via Let's Encrypt.

### 5a. Point DNS to the server

```
# At your registrar:
A    visitwesternnevada.org    -> <droplet-ip>
A    www.visitwesternnevada.org -> <droplet-ip>

# Wait 5-30 min, then verify from your laptop:
dig visitwesternnevada.org
```

### 5b. nginx config

```
sudo tee /etc/nginx/sites-available/ncexp > /dev/null <<'EOF'
server {
    listen 80;
    server_name visitwesternnevada.org www.visitwesternnevada.org;

    location / {
        proxy_pass http://127.0.0.1:5000;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }

    # Cache static assets aggressively
```

```
location ~* \.(jpg|jpeg|png|webp|gif|svg|css|js|woff2|ico)$ {
    proxy_pass http://127.0.0.1:5000;
    expires 7d;
    add_header Cache-Control "public, immutable";
}
}
EOF
```

```
sudo ln -s /etc/nginx/sites-available/ncexp /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default
sudo nginx -t
sudo systemctl reload nginx
```

## 5c. Add HTTPS

```
sudo certbot --nginx -d visitwesternnevada.org -d www.visitwesternnevada.org
# Enter email, agree to TOS, choose "redirect HTTP to HTTPS"
```

Certbot adds a cron job that auto-renews certificates 30 days before expiry. Set and forget.

## Phase 6 — Schedule the scraper

Nightly scrape at 3 AM, AI categorization 30 minutes later. Chamber staff review pending events the next morning.

```
# Edit the user's crontab
crontab -e

# Add these two lines:
0 3 * * * cd /home/ncexp/nevada-county-experience && \
    .venv/bin/python scraper/event_scraper.py >> /var/log/ncexp-scrape.log 2>&1
30 3 * * * cd /home/ncexp/nevada-county-experience && \
    .venv/bin/python scraper/ai_categorize.py >> /var/log/ncexp-ai.log 2>&1

# Make the log files writable
sudo touch /var/log/ncexp-scrape.log /var/log/ncexp-ai.log
sudo chown ncexp:ncexp /var/log/ncexp-scrape.log /var/log/ncexp-ai.log

# Run the scraper once manually to verify Selenium + Chromium work:
cd /home/ncexp/nevada-county-experience
.venv/bin/python scraper/event_scraper.py --site "Nevada City Chamber"
```

## Phase 7 — Backups

Two layers: provider snapshots for full-system recovery, plus a daily JSON dump for granular event/itinerary recovery.

### 7a. Provider snapshots (\$1/mo on DigitalOcean)

DO Console → Droplet → Backups → Enable. Weekly automatic, 4-week retention.

### 7b. Daily JSON backup

```
mkdir -p /home/ncexp/backups
cat > /home/ncexp/backup.sh <<'EOF'
#!/bin/bash
DATE=$(date +%Y%m%d)
cd /home/ncexp/nevada-county-experience
tar czf /home/ncexp/backups/ncexp-$DATE.tar.gz \
    scraper_output/events.json scraper/sources.json scraper/config.py
# Keep only last 30 days
find /home/ncexp/backups/ -name 'ncexp-*.tar.gz' -mtime +30 -delete
EOF
chmod +x /home/ncexp/backup.sh

# Schedule daily at 4 AM (after scrape + AI complete)
crontab -e
# Add:
0 4 * * * /home/ncexp/backup.sh
```

### 7c. Off-site copy (optional but recommended)

Once a month, copy the latest backup tarball to chamber NAS, Google Drive, or another cloud. Even a manual quarterly download protects against catastrophic provider loss.

## Phase 8 — Test & cutover

| Test                             | How to verify                                                                                     |
|----------------------------------|---------------------------------------------------------------------------------------------------|
| Public site loads at https://... | Browse to the domain. Confirm padlock + "https" + green/normal cert.                              |
| All vibes show experiences       | Click each of the 9 vibe cards; each should return matches.                                       |
| Events tab populates             | Should show ~460 approved events. If empty, server isn't reaching events.json.                    |
| Date filter works                | Pick "This Weekend" — events filter; some experiences hide based on season.                       |
| Mobile share works               | On a phone, build a list, hit Share. Native share sheet should appear.                            |
| Admin panel accessible           | Click ■ Manage Database. All four tabs should render.                                             |
| Scraper runs by hand             | <code>sudo -u ncexp .venv/bin/python scraper/event_scraper.py --site "Nevada City Chamber"</code> |
| AI Categorize button works       | Manage Database → Events Queue → ■ AI Categorize → enter 3 → confirm enrichment appears in ev     |
| Cron jobs scheduled              | <code>sudo crontab -u ncexp -l</code> should show the 3 AM and 3:30 AM lines.                     |
| Backups running                  | Wait until tomorrow morning, confirm <code>/home/ncexp/backups</code> has a new file.             |
| Service restarts                 | <code>sudo systemctl restart ncexp</code> ; site comes back within 2 seconds.                     |
| HTTPS auto-renews                | <code>sudo certbot renew --dry-run</code> should report success on all certs.                     |

## Phase 9 — Handoff to chamber

After the technical setup, hand the chamber team a small kit so they can operate the system day-to-day.

- **Admin URL bookmarked** on the chamber's computers — they only need to remember the public URL; the admin panel is reached from a button on the home page
- **Anthropic billing alerts** set up to email the chamber's ops manager at 50% / 75% / 100% of monthly cap
- **One-page cheat sheet** covering: how to approve events, how to add a venue, how to revoke an event after approval, how to trigger a manual scrape
- **Emergency contact** identified — who fixes it if the server is down? County IT? A contractor on retainer? The original developer?
- **Annual review on the calendar** — once a year, walk through all 161 experiences, verify hours/URLs/photos, prune anything closed
- **Backup test** — once before going live, restore from a backup to a test droplet to confirm the recovery process works
- **Domain renewal calendar reminder** — set 90 days before expiry so it never lapses

### Estimated effort & timeline

| Path                  | Time to live               | Approximate cost  |
|-----------------------|----------------------------|-------------------|
| Confident Linux admin | 4 hours (single afternoon) | ~\$400 contracted |

| Path                       | Time to live                | Approximate cost |
|----------------------------|-----------------------------|------------------|
| Self-taught learner        | 2 days (with research time) | time only        |
| County IT department       | Schedule-dependent          | internal         |
| Hosted-by-developer option | Same day                    | ~\$50/mo managed |

## Common pitfalls to watch for

- **Selenium / ChromeDriver version mismatch** — if scrapes fail with "session not created", apt usually keeps chromium-browser and chromium-chromedriver in sync, but a manual chromedriver download may be needed
- **Firewall too tight** — make sure ports 80 and 443 are open via ufw before running certbot, or HTTPS setup will fail
- **Time zone** — server defaults to UTC; cron jobs run in UTC. Either set the server to America/Los\_Angeles or schedule jobs in UTC (3 AM PT = 11 AM UTC)
- **File permissions on scraper output** — events.json must be writable by the ncexp user, or the scraper silently fails
- **Anthropic API key in repo** — confirm scraper/config.py is gitignored before first git push